Artificial Intelligence in Teaching and Teacher Professional Development: A Systematic Review

Abstract

The use of Artificial Intelligence (AI) technology in education is considered a major driver of educational innovation. While extensive literature exists on the integration of AI technologies in educational environments, the focus on teachers' roles and their professional development needs remains limited. This study presents a systematic review of research conducted between 2015 and 2024, analyzing how teachers are using AI technology in their teaching and professional development. Specifically, the study focuses on the relationship between the supply of professional development opportunities and the demand for AI integration by teachers. Utilizing PRISMA principles and protocols, this review identified and compiled a total of 95 significant research articles. The findings indicate a disproportionate focus of research. About 65% of the studies centered on the application of AI in teaching, including conversational AI, AI-based learning and assessment systems, immersive technologies, visual and auditory computing, and teaching and learning analytics. In contrast, only 35% of the studies explored the role of AI in teacher professional development. This review highlights the lack of sufficient research on integrating AI technologies into teaching practices, keeping in mind the professional development needs of teachers. It emphasizes the need for future research specifically studying how AI can be used to empower teacher professional development. Moreover, technical

and ethical challenges in AI-based professional development should be prioritized to ensure responsible and effective AI integration in education.